System Design Primer

Scalability, Caching, Databases, Load Balancing

1. Scalability Basics

- Vertical scaling: bigger machine (limited, single point of failure).
- Horizontal scaling: more machines behind a load balancer.

2. Load Balancing

- Algorithms: Round Robin, Least Connections, IP Hash.
- L4 (transport) vs L7 (application) load balancers.

3. Caching

- Cache-aside, Write-through, Write-back strategies.
- Redis/Memcached for in-memory caching.
- Cache eviction: LRU (least recently used), TTL.

4. Databases

- SQL vs NoSQL: structured/ACID vs flexible/schema-less.
- Indexes speed up reads, slow down writes.
- Sharding: horizontal partition of data by key.
- Replication: master-slave, leaderless.
- CAP theorem: pick 2 of Consistency, Availability, Partition tolerance.

5. Messaging & Async

- Queues decouple services (RabbitMQ, Kafka, SQS).
- Event-driven vs request-response architecture.

6. Key Interview Flow

- Clarify requirements -> estimate scale -> high-level design
- -> dive into components -> discuss trade-offs.
- Always consider: caching, load, DB choice, failure modes.